

Section 8 — Aliases & Computed Fields

Concept Notes

Sometimes the field names used by an external API or database are different from the names you want to use inside Python code.

Pydantic supports **aliases**, which allow one name to be used for incoming or outgoing data while keeping a different field name inside the model.

Pydantic also supports **computed fields**, which are values calculated from other fields instead of being provided directly by the user.

Field Alias

An alias gives a field an alternative external name.

For example, external JSON may contain:

userName

while Python code prefers:

username

An alias allows Pydantic to connect these two names.

Validation Alias

A validation alias controls which external key is accepted when data enters the model.

It is useful when input data uses a different naming style or comes from an external system whose field names cannot be changed.

Serialization Alias

A serialization alias controls the field name used when model data is exported.

For example, Python may use `product_name` internally, while an API response needs `productName`.

Computed Fields

A computed field is calculated from other model fields.

For example:

- `total_price = price × quantity`
- `full_name = first_name + last_name`
- `discount_amount = original_price - sale_price`

The computed value does not need to be provided manually.

In Pydantic V2, `@computed_field` is used to include calculated properties in serialized model output.

Important Points

- Aliases connect external field names with internal Python field names.
- alias can define an alternative field name.
- `validation_alias` controls the name accepted during validation.
- `serialization_alias` controls the name used during serialization.
- `by_alias=True` exports data using aliases.
- Aliases are useful when API naming conventions differ from Python conventions.
- Computed fields are calculated from other model fields.
- `@computed_field` is used for computed output fields.
- Computed fields reduce duplicate or manually maintained data.
- Computed fields are included in serialization output.

Syntax

Basic Alias

```
Field(  
    alias = "external_name"  
)
```

Validation Alias

```
Field(  
    validation_alias = "input_name"  
)
```

Serialization Alias

```
Field(  
    serialization_alias = "output_name"  
)
```

Export Using Alias

```
model_dump(  
    by_alias = True  
)
```

Computed Field

```
@computed_field  
@property
```

Create method



Calculate value



Return computed result